

PMGI SF5 / S1813 Bilayer Lift-Off Protocol

With DMSO Lift-Off for EGOFET Electrode Patterning

e-MolMat Group – ICMAB-CSIC
Protocol v1.1 - September 21, 2026

Abstract

This document describes an improved lift-off protocol for patterning Cr/Au source-drain and gate electrodes on Kapton substrates. The method replaces the standard single-layer Shipley S1813 process with a **PMGI SF5 / S1813 bilayer** stack, combined with a **DMSO-based lift-off** replacing the acetone/ultrasonic process. The bilayer creates a controlled undercut profile during development that prevents metal bridging on resist sidewalls, eliminating the need for manual cotton-swab removal of residual gold from transistor channels.

Contents

1	Introduction	2
1.1	Problem Statement	2
1.2	Proposed Solution	2
1.3	Required Materials	3
2	Process Overview	3
3	Detailed Process Steps	3
3.1	Substrate Preparation and Mounting	3
3.2	PMGI SF5 Coating and Prebake	4
3.3	S1813 Coating and Soft Bake	5
3.4	Exposure	5
3.5	Development (Single-Step Bilayer)	5
3.6	Cr/Au Evaporation	6
3.7	DMSO Lift-Off	7
3.8	Post-Lift-Off Cleaning	7
4	Troubleshooting	8
5	Process Comparison	9

1 Introduction

1.1 Problem Statement

The current EGOFET fabrication protocol uses a single-layer Shipley S1813 positive photoresist for lift-off patterning of Cr/Au electrodes. This produces vertical or slightly positive sidewall profiles, causing the evaporated metal to form a continuous film on the resist sidewalls. During lift-off, the acetone/ultrasonic baths cannot fully dissolve the resist underneath this metal “fence,” leaving gold residue in the Source–Drain channels. This residue must be removed manually with acetone-soaked cotton swabs, which frequently damages the electrode pattern.

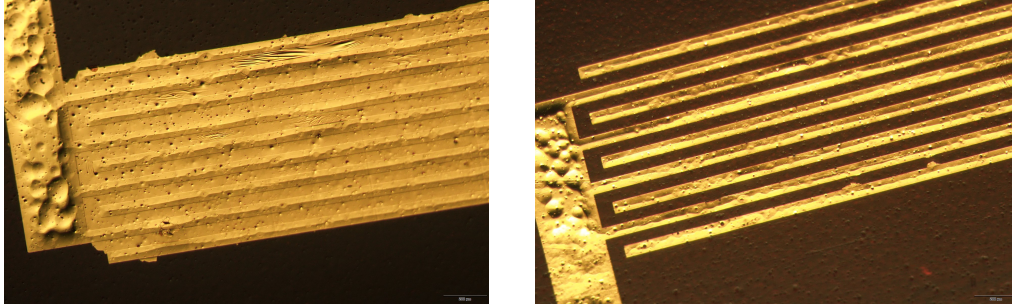


Figure 1: Left: Gold residue in Source–Drain channel after lift-off process using single-layer S1813. Right: Clean channel after manual cotton-swab removal. This can lead to pattern damage.

1.2 Proposed Solution

A bilayer resist stack consisting of:

- **PMGI SF5** (bottom layer): A polydimethylglutarimide-based lift-off resist that dissolves isotropically in alkaline developer, creating a controlled undercut beneath the imaging resist.
- **Shipley S1813** (top layer): The standard positive imaging resist, processed identically to the current protocol.

The bilayer stack is represented in Figure 2.

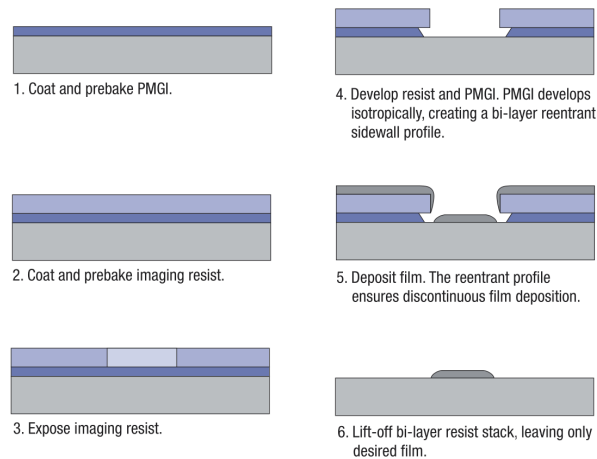


Figure 2: Schematic of the bilayer lift-off process.

The undercut profile ensures that the evaporated metal film is discontinuous at the resist edges, allowing the solvent to penetrate and lift off the unwanted metal cleanly. DMSO at 60 °C replaces acetone/ultrasonic for a gentler, more effective strip.

1.3 Required Materials

Table 1: Materials and chemicals required.

Material	Specification	Status
PMGI SF5	MicroChem lift-off resist	In stock
Shipley S1813	Positive photoresist	In stock
MF-319	TMAH developer (2.2%)	In stock
DMSO	Dimethyl sulfoxide, ULSI grade	Purchased
IPA	Isopropanol, cleanroom grade	In stock
DI water	18.2 M Ω · cm	Available
Kapton HN	75 μ m polyimide foil	In stock
Si carrier wafer	4-inch, for mounting	Available

2 Process Overview

The bilayer lift-off process consists of eight major steps:

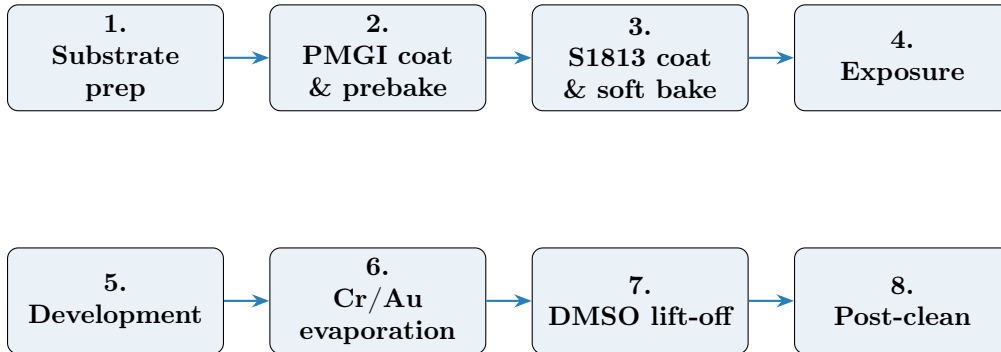


Figure 3: Bilayer lift-off process overview.

3 Detailed Process Steps

3.1 Substrate Preparation and Mounting

- a) Cut Kapton foil to 8 cm $\times$ 8 cm pieces. Allow cut pieces to flatten on a clean surface for 2–3 days before use to avoid ridges.
- b) Clean the Kapton substrate:
 - i) Rinse with acetone
 - ii) Rinse with isopropanol
 - iii) Dry thoroughly with N₂ gun (front and back)
- c) Mount the Kapton on a clean Si carrier wafer using the **IPA capillary method**:
 - i) Place the first Kapton tape strip on one edge of the foil.

- ii) Dispense a small drop of IPA between the Kapton foil and the Si carrier. The IPA wicks by capillarity, pulling the foil flat against the carrier.
 - iii) Press gently to adhere.
 - iv) Apply tape on the opposite edge, pulling the foil taut without stretching.
 - v) Apply tape on the remaining two edges.
 - vi) Leave **small gaps** between tape strips for air evacuation during evaporation.
- d) Ensure both Kapton and carrier are **completely dry** — trapped humidity at the interface ruins exposure and evaporation. Verify flatness by visual inspection: no wrinkles, no bubbles, no lifting corners.

Important

Use **two different Si carriers**: one for photolithography (coating + exposure + development) and a fresh one for evaporation. This prevents adhesive residue from contaminating the evaporator.

3.2 PMGI SF5 Coating and Prebake

PMGI SF5 Parameters

Parameter	Value
Spin speed	3000 rev min ⁻¹ (50 Hz)
Spin time	45 s
Acceleration	4000 rev min ⁻¹ s ⁻¹
Expected thickness	0.5 μm to 0.7 μm
Prebake temperature	170 °C
Prebake time	5 min
Bake mode	Contact hotplate

- a) Clean the spin coater chuck and support with acetone.
- b) Place a square of double-sided tape under the carrier wafer to prevent movement during spinning.
- c) Apply PMGI SF5 by pouring directly from the bottle (avoid pipette to prevent bubbles). Cover the wafer surface completely.
- d) Press START immediately after dispensing.
- e) Transfer to hotplate for prebake at 170 °C for 5 min.

Tip: Prebake Temperature

The prebake temperature is the primary control for undercut rate. At 170 °C with MF-319, the undercut rate is approximately 33 Å s⁻¹ (0.33 nm s⁻¹). This provides a good balance between process control and development speed.

If undercut is too aggressive (features lifting off), increase prebake to 190 °C (28 Å s⁻¹). If undercut is insufficient (metal residue remains), decrease to 150 °C (42 Å s⁻¹).

3.3 S1813 Coating and Soft Bake

S1813 Parameters

Parameter	Value
Spin speed	5000 rev min ⁻¹ (83.33 Hz)
Spin time	25 s
Acceleration	4000 rev min ⁻¹ s ⁻¹
Expected thickness	1.3 μm
Soft bake temperature	75 °C
Soft bake time	1 min
Bake mode	Contact hotplate

- Apply S1813 directly on top of the baked PMGI layer. **No intermixing occurs** — PMGI is insoluble in the ethyl lactate solvent of S1813.
- Spin at 5000 rev min⁻¹ for 25 s.
- Soft bake at 75 °C for 1 min.

Do Not

Do not press manually on the spinner lid during coating. Do not use HMDS — PMGI has excellent adhesion without primers.

3.4 Exposure

- Expose the pattern using the Microwriter ML3 laser direct-write system.
- Use the **same dose** as the current S1813-only process.
- Recommended focus calibration: 9 points with offsets (0, 0), (± 30 , 0), (0, ± 30), (± 20 , ± 20) μm .

3.5 Development (Single-Step Bilayer)

Development Parameters

Parameter	Value
Developer	MF-319 (TMAH 2.2% with surfactant)
Mode	Immersion, manual stirring
Time	60 s to 90 s
Rinse	DI water, ≥ 300 mL
Dry	N ₂ gun, front and back

- Carefully remove the Kapton foil from its Si carrier (do not bend).
- Immerse in MF-319 with the patterned face **up**.
- Stir manually for 60 s to 90 s.
- Rinse thoroughly with DI water (300 mL or more).
- Dry **front and back** with N₂ gun — residual water affects evaporation quality.

Understanding the Development

MF-319 develops **both layers simultaneously**:

- The exposed S1813 dissolves in the developer (positive-tone behavior).
- The underlying PMGI dissolves **isotropically** — sideways as well as downward — creating the undercut.

The undercut depth is:

$$d_{\text{undercut}} = r_{\text{PMGI}} \times t_{\text{develop}}$$

where $r_{\text{PMGI}} \approx 33 \text{ Å s}^{-1}$ at 170°C prebake with MF-319.

For 60 s: $d \approx 2.0 \text{ μm}$ of undercut.

For 90 s: $d \approx 3.0 \text{ μm}$ of undercut.

Both values are sufficient for 45 nm of metal deposition.

Critical

Do **not** over-develop. Excessive undercut can cause small features ($<10 \text{ μm}$) to detach from the substrate. Start with 60 s and increase only if metal residue persists after lift-off.

3.6 Cr/Au Evaporation

- Re-mount the developed foil on a **fresh** Si carrier (different from the one used for photolithography).
- Provide the evaporator technician with the following parameters:

Evaporation Parameters — Request to Technician

Parameter	Value
Adhesion layer	Cr, 5 nm
Electrode layer	Au, 40 nm
Cr deposition rate	$\leq 0.2 \text{ Å s}^{-1}$
Au deposition rate	$\leq 0.5 \text{ Å s}^{-1}$
Base pressure	$\leq 5 \times 10^{-6} \text{ mbar}$
Substrate rotation	None (incidence normal)

Quality Check

After evaporation, the Au film should appear **gold-colored and reflective**. If it looks pinkish, opaque, or matte, inform the technician — this indicates oxidation or contamination and will cause poor SAM formation and device performance.

3.7 DMSO Lift-Off

DMSO Lift-Off Parameters

Parameter	Value
Solvent	DMSO (dimethyl sulfoxide), ULSI grade
Temperature	60 °C
Duration	2 h to 3 h
Ultrasonic	None (passive soak only)
Agitation	Gentle manual swirling every 30 min

- Place the sample in a glass beaker with fresh DMSO.
- Cover with a watch glass or petri dish (DMSO has low vapor pressure but should still be contained).
- Place on hotplate at 60 °C.
- Soak for 2 h to 4 h. Swirl gently every 30 min.
- The metal “skin” on top of the resist will begin to detach and float to the surface.
- Once lift-off is complete, remove sample and rinse immediately with IPA.

Why DMSO, not Acetone?

DMSO is preferred over acetone for the following reasons:

- **No ultrasonic required:** The undercut profile allows DMSO to penetrate under the metal without mechanical agitation, preventing damage to delicate electrode patterns.
- **Better penetration:** DMSO has lower surface tension and higher boiling point than acetone, allowing it to wick into the undercut gap more effectively.
- **Non-toxic:** DMSO is not classified as toxic, unlike NMP (the traditional lift-off solvent).
- **Compatible with Kapton:** DMSO does not attack polyimide substrates.

3.8 Post-Lift-Off Cleaning

- Rinse with IPA: **3 baths** × 15 min each, fresh IPA per bath. The additional bath ensures thorough removal of DMSO residues from the gold surface.
- Dry with N₂ gun.
- Inspect under microscope: verify clean channels with no residual gold.

DMSO–Au Surface Interaction

DMSO coordinates to gold surfaces through its sulfur atom ($\text{S}-\text{O} \rightarrow \text{Au}$), forming molecular complexes that can modify the Au work function and interfere with subsequent SAM formation. The triple IPA rinse followed by the UV-ozone treatment (25 min) in the SAM preparation step effectively removes these organic residues and restores the Au surface. **Always perform SAM immediately after UV-ozone activation.**

Expected Result

After successful lift-off, gold should be present **only** in the patterned electrode areas. The Source–Drain channels should be **completely clean** — no gold islands, no metal fences, no need for cotton-swab removal. If residue persists, see Section 4.

4 Troubleshooting

Table 2: Common issues and solutions.

Symptom	Likely Cause	Solution
Gold residue in channels	Insufficient undercut	Increase develop time by 15 s increments. Check PMGI pre-bake temperature.
Electrode pattern de-taching	Excessive undercut	Decrease develop time or increase PMGI prebake temperature to 190 °C.
Lift-off takes >3 h	DMSO too cold or undercut too small	Verify hotplate temperature. Increase develop time.
Pinkish Au film	Evaporator contamination	Re-evaporate. Inform technician.
Poor SAM formation	Au surface contaminated	Perform UV-ozone clean (25 min) immediately before SAM.
PMGI not adhering to Kapton	Surface contamination	Ensure dehydration bake at 200 °C/5 min before PMGI coating.
Ridges/hills in Au film	Kapton not flat on carrier	Use IPA capillary method for mounting (Section 3.1). Ensure foil is pre-flattened for 2–3 days. Verify no wrinkles before coating.

5 Process Comparison

Table 3: Comparison: Current protocol vs. Bilayer protocol.

Parameter	Current (S1813 only)	Bilayer (PMGI + S1813)
Photoresist layers	1 (S1813)	2 (PMGI SF5 + S1813)
Sidewall profile	Vertical	Undercut (re-entrant)
Lift-off solvent	Acetone + ultrasonido	DMSO at 60 °C
Lift-off time	3 × 15 min + IPA baths	2 h to 3 h passive
IPA rinse baths	2 × 15 min	3 × 15 min
Cotton swab needed	Yes (frequently)	Rarely (<42%)
Pattern damage risk	High	Low
Developer	MF-319	MF-319 (same)
Exposure tool	Microwriter ML3	Microwriter ML3 (same)
Kapton mounting	Tape strips	IPA capillary method
Additional equipment	None	Hotplate capable of 170 °C
New chemicals	None	DMSO only

References

- [1] MicroChem Corp., “PMGI Resists — technical data sheet,” tech. rep., Kayaku Advanced Materials, 2021. Available: <https://kayakuam.com/product/structure/lift-off-resists>.
- [2] MicroChem Corp., “LOR Lift-Off Resists — technical data sheet,” tech. rep., Kayaku Advanced Materials, 2021.
- [3] MicroChemicals GmbH, “DMSO (ULSI) — product information,” 2026. Available: <https://www.microchemicals.com>.
- [4] Nanyte, “Lift-off lithography, step by step,” 2026. Tutorial. Available: <https://nanyte.com/photoresist-tutorials/lift-off>.
- [5] F. P. Bonafe *et al.*, “Dissipative charge transport in organic mixed ionic-electronic conductor channels,” *Nature Communications*, vol. 16, p. 4537, 2025.
- [6] S. Ruiz-Molina, *EGOFETs for biosensing*. PhD thesis, ICMAB-CSIC, 2024.
- [7] S. Ricci, *Organic electrochemical transistors for biosensing*. PhD thesis, ICMAB-CSIC, 2020.
- [8] Fraunhofer IPMS, “OFET Substrates — test chips for material characterization,” 2026. Protocol for resist removal.
- [9] University of Minnesota Nano Center, “PMMA/PMGI Bilayer E-Beam Liftoff Process,” 2026. Institutional protocol. Available: <https://cse.umn.edu/mnc>.